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Chapter 1

Results and Discussion

Following the experimental and analysis procedures explained in Chapter 2, the ODT capture efficiency was characterized and optimized. This chapter presents the corresponding experimental results.

The first stage of the optimization process consists of minimizing the molasses temperature through the adjustment of the cooling laser detuning, the cooling laser intensity, and the compensation magnetic field.

Subsequently, the ODT loading is optimized by exploiting the compensation coils to displace the MOT after loading. This optimization aims to maximize the spatial overlap between the atomic cloud and the ODT beam, thereby enhancing the capture efficiency.

1.1 Molasses Temperature Optimization

Before starting the optimization, the six MOT beams were realigned to remove any residual overlap mismatch at the trap position — a necessary precaution, since thermal drifts in the optical components tend to slowly spoil this alignment over time. The compensation coil currents were initially set to the values found in earlier characterizations of the apparatus,

$$I_x = 80 \text{ mA}, \quad I_y = -200 \text{ mA}, \quad I_z = -400 \text{ mA}, \quad (1.1)$$

which served as the starting point for the systematic optimization described below.

1.1.1 Detuning Frequency Optimization

The optimization began with the cooling-laser detuning, scanned from -24.4 MHz ($4.0 \Gamma/2\pi$) to -184 MHz ($30.4 \Gamma/2\pi$) in steps of -11.4 MHz , and the molasses temperature measured at each point.

At small detunings the measured temperature closely tracks the $1/|\delta|$ dependence expected for sub-Doppler (Sisyphus) cooling. This trend breaks down as the detuning grows further: the scattering rate falls off as $\Gamma_p \propto 1/\delta^2$, and the detuning approaches the $F = 2 \rightarrow F' = 2$ transition, only 267 MHz below the main cooling transition $F = 2 \rightarrow F' = 3$. Both effects flatten the temperature curve in the range $|\delta| \approx 15\text{--}20$ and eventually turn it upward again at larger detunings.

The lowest temperature was found at a normalized detuning of $|\delta| = 17.2$, i.e. 104.2 MHz in absolute terms. Fitting the ballistic expansion at this detuning gave $T_x = (14.1 \pm 0.5) \mu\text{K}$ and $T_y = (13.5 \pm 0.7) \mu\text{K}$ along the two axes, whose weighted average yields a final temperature of $T = (13.9 \pm 0.4) \mu\text{K}$.

The fit at this detuning shows a systematic feature common to all such measurements: the data points at the longest times of flight fall consistently below the linear trend expected from the expansion law. The most likely explanation is that, at sufficiently long expansion times, the cloud radius is no longer small compared to the probe beam waist, so the assumption of uniform illumination breaks down and the outer, dimmer part of the cloud is under-weighted in the fit — an effect that systematically biases the extracted variance downward. Fortunately, this bias is expected to become less severe as the temperature (and hence the expansion rate) decreases further.

1.1.2 AOM Amplitude Optimization

The next optimization step examined the influence of the cooling-beam power itself, controlled via the RF drive amplitude of the switching AOM, together with the duration of the ramp used to transition this amplitude from its MOT value down to the molasses value at the start of the molasses phase. Three amplitude settings were tested — 385, 475, and 550 a.u., corresponding to 5.8, 6.6, and 6.3 mW of cooling power measured after the z -axis MOT collimator (the MOT itself is operated at 550 a.u.) — combined with ramp durations of either 5 ms or 20 ms, and the whole set was repeated across several values of the cooling detuning.

No statistically significant dependence of the molasses temperature on the AOM amplitude emerged from this scan — the small variations observed between the three settings remained within the experimental uncertainty, so no clear optimal amplitude could be identified in the tested range. The ramp duration, on the other hand, had a small but systematic effect: the 20 ms ramp consistently gave temperatures about $0.5 \mu\text{K}$ lower than the 5 ms ramp. Given that this improvement was already small and no further gain at the level of a few tenths of a microkelvin was required for the purposes of this work, no additional ramp durations were tested, and the 20 ms ramp was adopted for the subsequent measurements.

It is worth noting, anticipating the ODT characterization described later in this chap-

ter, that this particular choice was later found to introduce extra fluctuations once the ODT loading was being optimized. For that reason, the compensation-coil optimization described next was actually carried out with the AOM amplitude set to 385 a.u. — the setting that gave the lowest temperature at $|\delta| = 17.2 \Gamma/2\pi$ — together with the 20 ms ramp, even though both parameters were revisited again during the ODT optimization stage. As shown by the measurements in this section, however, their influence on the molasses temperature itself is at most a few tenths of a microkelvin, so this later readjustment does not affect the temperature results reported here.

1.1.3 Compensation Field Optimization

The last step of the molasses optimization concerned the compensation (shim) magnetic field, whose currents are switched from their MOT values to their molasses values 5 ms after the MOT quadrupole field itself is turned off. The three axes were optimized sequentially: first the z -axis current, with the x - and y -axis currents held at their preliminary values; then the x -axis current, using the newly found optimal z value; and finally the y -axis current, with both z and x set to their respective optima. At each step, a parabolic fit to the measured temperature versus current was used to extract the optimal current and the corresponding minimum temperature.

The z -axis scan gave a minimum temperature of $(11.46 \pm 0.03) \mu\text{K}$ at $I_z = (-223.2 \pm 0.7) \text{ mA}$; the subsequent x -axis scan, performed at this z current, gave $(9.2 \pm 0.1) \mu\text{K}$ at $I_x = (-17 \pm 2) \text{ mA}$; and the final y -axis scan gave $(9.6 \pm 0.1) \mu\text{K}$ at $I_y = (-184 \pm 3) \text{ mA}$. The currents actually adopted for the subsequent ODT loading stage were

$$I_x = -15 \text{ mA}, \quad I_y = -185 \text{ mA}, \quad I_z = -230 \text{ mA}, \quad (1.2)$$

corresponding, via the coil calibration given in Eq. (??), to compensation fields of

$$B_x = -21 \text{ mG}, \quad B_y = -241 \text{ mG}, \quad B_z = -345 \text{ mG}. \quad (1.3)$$

With this compensation field, the molasses temperature reached its minimized value of $T = (9.6 \pm 0.1) \mu\text{K}$. Comparing the three axes, the z -axis current required the largest correction from its preliminary value, the x -axis current a smaller but still appreciable one, while the y -axis current turned out to already be close to its optimal value beforehand. The operating current actually used for z , -230 mA , differs slightly from the true fitted optimum of -223.2 mA ; since both values sit close to the vertex of the fitted parabola, where the temperature is only weakly sensitive to current, this small offset does not introduce any appreciable systematic error in the reported temperatures.

1.2 ODT Loading Optimization

With the molasses parameters established in the previous section, the next stage of the optimization targets the ODT loading itself, primarily by tuning the compensation magnetic field. The idea behind this step is that the MOT forms at the zero of the magneto-optical potential, whose location can be shifted by adjusting the compensation coil currents; displacing the trapped cloud in this way, immediately before the molasses phase, allows its position to be matched to that of the ODT beam. Moving the MOT away from the geometric center where the six cooling beams overlap best inevitably costs some atoms and raises the temperature slightly, but this trade-off is worthwhile if it improves the spatial overlap with the ODT and thereby increases the number of atoms actually transferred into the dipole trap. As long as the displacement stays small, the molasses parameters optimized earlier remain effective at cancelling the residual magnetic field and require no further adjustment.

In practice, once the MOT has loaded, the compensation coil currents are ramped smoothly to their ODT-loading values over 200 ms, discretized into 20 steps, so as to minimize atom loss and heating during the displacement. As with the molasses optimization, preliminary current values — obtained before the beam realignment and molasses optimization described above — were available as a starting point,

$$I_x = -200 \text{ mA}, \quad I_y = -350 \text{ mA}, \quad I_z = 50 \text{ mA}. \quad (1.4)$$

Throughout these measurements, the ODT laser power before injection into the fiber was 1.4 W. The transmission efficiency up to just before lens L3 (see the apparatus schematic in the previous chapter) was measured at low power to be about 42%, corresponding to roughly 0.59 W; since several optics sit upstream of this measurement point, the actual power reaching the fiber facet is likely somewhat higher, so a conservative 10% relative uncertainty was assigned, giving an estimated fiber-facet power of $P = 0.59 \pm 0.06 \text{ W}$.

1.2.1 Imaging System Optimization

The first attempts to characterize the ODT capture efficiency used the FLIR camera, with the molasses parameters already optimized above and the preliminary compensation currents quoted for the ODT stage. The analysis followed the procedure described in the previous chapter, but without the background rescaling step.

With the 20 ms AOM ramp used for the molasses optimization, the resulting 2D difference image showed no distinct bright region attributable to the ODT, and the 1D integrated profile only hinted at a weak, localized increase spanning about 4 pixels around the fiber position — later confirmed, from subsequent measurements, to indeed correspond to the

trapped atoms, but too small to be reliably distinguished from noise. Since integrating along one dimension is specifically meant to boost the signal-to-noise ratio, this persistent weak signal pointed to a real underlying problem rather than a simple statistical limitation: the residual background pedestal left after subtraction was significantly larger than the ODT signal itself, suggesting oscillatory or shot-to-shot correlated noise that five-image averaging alone could not remove. In addition, the ODT signal spanned only a handful of pixels, which is both consistent with a lower-than-expected initial capture (even accounting for a non-optimal cloud-ODT overlap) and, independently, an obstacle to analysis: too few illuminated pixels to support a reliable 2D Gaussian fit, while direct pixel summation would introduce its own systematic errors.

To address the background problem, the molasses parameters themselves were revisited, and the AOM ramp duration was found to be the key culprit: shortening it from 20 ms to 5 ms markedly improved the background subtraction and thus the signal-to-noise ratio. At the same time, the AOM amplitude was set to 480 a.u. (6.6 mW of cooling power on the z -beam), a value chosen because it sits in the saturated part of the AOM diffraction-efficiency curve, where $dP/dA \approx 0$; operating at this point intrinsically suppresses the sensitivity of the delivered optical power to amplitude noise in the AOM drive.

With this shorter ramp, a distinct bright strip corresponding to the ODT-trapped atoms became visible in the 2D difference image, and comparing the 1D profiles obtained with the two ramp durations shows that the residual pedestal is suppressed to roughly half its previous amplitude and area with the 5 ms ramp — enough for the ODT signal to clearly emerge above the background.

Even with this improved contrast, the FLIR camera’s spatial resolution remained insufficient. The subsequent measurements therefore switched to the Manta camera, whose smaller pixel pitch was further complemented by mounting it closer to the chamber, at the minimum working distance of the MVL50M23 objective, to maximize magnification. Comparing the apparent width of the fiber as imaged by the two cameras indicated that this new configuration increased the object size in pixels by a factor of about 2.25 — the combined effect of the higher magnification and the smaller pixel size — which finally provided a sufficiently well-resolved ODT signal for robust quantitative analysis.

1.2.2 Capture Efficiency Optimization

With adequate imaging in place, the compensation coil currents were optimized systematically using the full analysis procedure from the previous chapter, this time with the Manta camera. The three axes were scanned sequentially in the order $x \rightarrow z \rightarrow y$, with the currents on the not-yet-optimized axes held at the preliminary values quoted above. Both the ODT signal area — directly proportional to the number of captured atoms, though only measurable in arbitrary units since the camera was not calibrated for absolute

atom counting — and the capture efficiency extracted from these scans showed matching trends along all three axes, as expected since they are two ways of looking at the same underlying signal. This first pass yielded the optimal currents

$$I_x = -220 \text{ mA}, \quad I_y = -290 \text{ mA}, \quad I_z = -90 \text{ mA}. \quad (1.5)$$

To confirm this result, the z -axis scan was repeated, and this second run revealed an unexpected feature.

Rather than the expected single-peaked, roughly parabolic dependence, both the ODT area and the capture efficiency showed an oscillatory behavior as a function of the z -axis current: starting from -1000 mA , the signal rose to a first local maximum near -800 mA , dipped around -700 mA , and rose again to further local maxima near -600 mA and -400 mA . The same periodicity appeared in the background area — the molasses cloud size measured with the ODT off at the end of the time of flight — with peaks at essentially the same current values, and again in the x -centroid of the cloud on the camera, an oscillation not seen when scanning the other two compensation axes. While a larger initial MOT population naturally leads to a larger captured ODT fraction, the fact that this covariation is periodic in current is not obviously explained; a plausible contributing factor is the complex spatial intensity distribution of the cooling beams near the trap center combined with residual misalignment, but confirming this would require dedicated further measurements, which were beyond the scope of this work.

Taking this behavior into account, the compensation currents and the corresponding generated field components were finally set to

$$I_x = -220 \text{ mA}, \quad I_y = -290 \text{ mA}, \quad I_z = -600 \text{ mA}, \quad B_x = -308 \text{ mG}, \quad B_y = -377 \text{ mG}, \quad B_z = -900 \text{ mG}, \quad (1.6)$$

Converting these compensation fields into an equivalent displacement of the MOT position via $t_\alpha = B_\alpha/B'$ for $\alpha = x, y$ and $t_z = -B_z/2B'$, with $2B' = 9.32 \text{ G/cm}$ the axial gradient of the quadrupole field, gives

$$t_x = -661 \mu\text{m}, \quad t_y = -809 \mu\text{m}, \quad t_z = 966 \mu\text{m}. \quad (1.7)$$

This configuration maximized the ODT trap area — and hence the number of captured atoms — and corresponds to a capture efficiency of $\eta = (0.160 \pm 0.005)\%$.

This measured efficiency can be compared with the shallow- and non-shallow-trap models introduced in the previous chapter, evaluated at the best molasses configuration found above: $w_0 = 19.0 \mu\text{m}$, $P = (0.59 \pm 0.06) \text{ W}$, $R_x R_y \approx (1.043 \pm 0.004) \text{ mm}^2$, and $T = (9.60 \pm 0.10) \mu\text{K}$. These give $\eta = (0.45 \pm 0.05)\%$ for the shallow-trap model and $\eta = (0.41 \pm 0.04)\%$ for the non-shallow-trap model; given $\xi = 0.39 \pm 0.04$, the close agreement between the

two models is expected. Note that, unlike in the earlier discussion of these models, no factor of $1/2$ for the fraction of atoms with $v_z < 0$ was applied here, since that factor was originally introduced to estimate only the atoms guided into the fiber core, whereas the present measurement targets the total number of captured atoms regardless of their velocity direction.

The measured efficiency is of the same order of magnitude as the theoretical prediction, indicating that the apparatus is operating correctly and close to its optimal parameters, but the theoretical value remains about 2.5 times larger than the measured one. Part of this discrepancy may be a genuine model limitation: the calculation assumes a purely Gaussian ODT beam, an excellent approximation for the fundamental LP_{01} mode, but the beam actually transmitted by the fiber also carries higher-order modes, as discussed in the previous chapter. Before attributing the full discrepancy to the model, however, a procedural effect should also be considered: the imaging sequence requires $600\ \mu\text{s}$ with the ODT switched off at the end of the time of flight, and atom loss during this dead time could artificially lower the measured trapping efficiency.

1.3 Simulation Results

1.3.1 Validation on the Ideal Three-Level System

Before addressing the full atomic structure, the cooling dynamics were first verified on the idealized three-level Λ system, for which the dark-state interference is exact by construction and the numerics are comparatively cheap to run with the exact master-equation solver.

The resulting animation shows the three expected features side by side: the vibrational population starting from a hot initial distribution ($n = 15$) and progressively piling up on the motional ground state; the steady-state Fano spectrum, with its dark transparency window sitting exactly on the carrier frequency and its narrow absorption peak aligned with the red motional sideband; and the corresponding decay of the mean phonon number $\langle n \rangle$ over time as the atom is cooled. This confirmed that the solver and the underlying EIT cooling mechanism behave as expected before the added complexity of the real atomic structure was introduced.

One caveat applies to this illustrative run: starting from $n = 15$ with $\eta = 0.35$ puts the system only marginally inside the Lamb–Dicke regime at $t = 0$ ($\eta\sqrt{\langle n \rangle + 1} \approx 1.4$), so the linearized motional coupling used in the interaction Hamiltonian mildly underestimates higher-order sideband transitions during the very earliest stage of the evolution. Since the population is rapidly cooled toward $\langle n \rangle \approx 0$, however, the system quickly settles deep into the Lamb–Dicke regime, so the computed steady-state cooling limit itself is unaffected by this initial-transient approximation.

1.3.2 Repumper Optimization

Within the full 24-level model, an accumulation of population in the uncoupled $F = 1$ ground-state manifold was observed during the cooling dynamics — atoms that fall out of the $F = 2$ cooling cycle and stop interacting with the probe and coupling lasers, introducing effective dead time that limits the achievable cooling rate. The repump Rabi frequency Ω_{repump} was therefore scanned, using the steady-state Fano spectrum and ground-state leakage populations as diagnostics.

The scan revealed a direct trade-off: a stronger repump field is more effective at depleting the $F = 1$ population, but it also dresses the atomic states more strongly, producing an AC Stark shift that displaces the entire Fano profile toward higher probe detuning. At low repump power ($\Omega_{\text{repump}} = 0.2\gamma$), population trapping in $F = 1$ reached about 35%, substantially isolating a third of the ensemble from the active cooling transition; increasing Ω_{repump} to 0.7γ and beyond was sufficient to deplete this residual population and restore the ensemble to the $F = 2 \leftrightarrow F' = 2$ cycle. At the same time, the induced

light shift grows with Ω_{repump} , and at $\Omega_{\text{repump}} = 0.7\gamma$ this shift was found to align the Fano absorption maximum precisely with the target red sideband; for higher values (e.g. 1.5γ or 2.0γ) the shift overshoots this alignment and the absorption peak moves past the sideband, decoupling the atom from the cooling transition. Based on this trade-off, $\Omega_{\text{repump}} = 0.7\gamma$ was identified as the optimal repump strength for this parameter set, balancing dark-state depopulation against excess AC Stark shift.

1.3.3 Probe Optimization

The same diagnostic procedure was then applied to the probe Rabi frequency Ω_p , scanned from 0.05γ to 0.35γ .

Here the limiting factor is power broadening: EIT cooling relies on a steep asymmetry between near-zero absorption at the carrier and strong absorption at the red sideband, and at low probe power ($\Omega_p = 0.05\text{--}0.10\gamma$) this contrast is well preserved. As Ω_p increases, the dark-resonance dip fills in and the absorption peak broadens, degrading the achievable contrast and, correspondingly, raising the expected carrier-scattering rate and the resulting steady-state temperature. A probe-induced AC Stark shift was also observed, distinct from the repumper-induced one: for Ω_p between 0.15γ and 0.35γ the absorption peak systematically drifts to higher detuning, eventually detaching from the target sideband at the highest powers tested. Increasing the probe power also degrades the steady-state ground-state preparation, by increasing the overall scattering rate and driving more population into $F = 1$ and into spurious $F = 2$ Zeeman sublevels, even though the fractional leakage into the off-resonant $F' = 3$ state actually decreases with probe power (since it is a relative measure against a growing target-transition signal). On balance, the weak-probe regime was found to be preferable, and $\Omega_p = 0.10\gamma$ was adopted as the operating point, preserving a sharp Fano feature aligned with the red sideband while keeping the excitation amplitude adequate.

1.3.4 Coupling Detuning Optimization

With the repumper and probe powers fixed, the coupling-laser detuning Δ_c was scanned from $+6.0\gamma$ to $+13.0\gamma$.

Because the coupling field itself induces an AC Stark shift, the position of the Fano absorption maximum drifts with Δ_c : at the lower end of the scanned range ($+6.0\text{--}+8.0\gamma$) the peak falls short of the target red sideband, while the absolute height of the peak decreases monotonically with Δ_c (from about 0.0008 at $+6.0\gamma$ down to about 0.0004 at $+13.0\gamma$), which on its own would favor smaller detunings for a larger excitation amplitude. The two effects were found to balance at $\Delta_c = +9.0\gamma$, where the absorption maximum aligns with the red sideband while retaining sufficient amplitude, and where the corresponding

minimum in the off-resonant $F' = 3$ leakage spectrum — marking the EIT transparency window — is properly positioned to suppress carrier scattering. This value was therefore adopted as the operating detuning for the subsequent time-dependent simulations.

1.3.5 Cooling Dynamics with the Optimized Parameters

Combining the optimized repumper, probe, and coupling-detuning values ($\Omega_{\text{repump}} = 0.7\gamma$, $\Omega_p = 0.10\gamma$, $\Delta_c = +9.0\gamma$), the full time-dependent cooling dynamics were computed for the 24-level configuration in which both Λ -system ground states lie within $F = 2$.

With this parameter set the simulated system cools in fewer than 10^5 simulation steps, corresponding to a cooling time of approximately 2 ms — a substantial improvement over the unoptimized configuration used during the initial parameter scans, whose cooling time was around 15 ms. Obtaining this Monte Carlo trajectory for the full 24-level system required about 8 hours of computation; for comparison, the exact master-equation solver applied to the same system only reached about 70,000 simulated steps after three days of continuous running, confirming that the Monte Carlo wavefunction method is the only practical option for the complete atomic structure.

1.3.6 Cooling Dynamics with Ground States Split Across $F = 1$ and $F = 2$

The same optimization methodology was then repeated for the alternative level assignment, in which the two Λ -system ground states are split between $F = 2$ and $F = 1$ rather than both lying in $F = 2$, and the repumper instead recovers population left behind in $F = 2$. This configuration is the one that more closely matches the level structure accessible in our experimental setup, and is therefore the one carried forward into the real-parameter simulation described below.

Applying the same procedure — steady-state Fano spectrum evaluation followed by sequential optimization of the laser parameters and a Monte Carlo time evolution — yielded the optimized parameters $\Delta_c = +9\gamma$, $\Omega_p = 0.17\gamma$, $\Omega_r = 1.27\gamma$ for this configuration. As in the previous configuration, the repumper was found to be essential: without it, population is rapidly lost to uncoupled dark states and the cooling dynamics stall.

This behavior was checked explicitly by running the simulation without the repump field, at zero magnetic field, and was found to hold equally in the presence of a magnetic field — in fact, the Fano profile itself is further degraded once Zeeman shifts are included, making the repumper’s role even more important in the presence of a realistic bias field.

1.3.7 Axial-Direction Simulation with Real System Parameters

As a final step, the $F = 1/F = 2$ -split configuration was re-evaluated using the trap frequency ν and Lamb–Dicke parameter η specific to the axial direction of our experimental trap geometry, in place of the generic illustrative values used in the parameter scans above. The same sequential optimization procedure was applied to this realistic configuration, converging again on parameters close to $\Delta_c = +9\gamma$, $\Omega_p = 0.17\gamma$, $\Omega_r = 1.27\gamma$, and the resulting parameter set was then used to run the full Monte Carlo time evolution for this specific geometry.

The resulting cooling dynamics for the axial direction of our actual trap follow the same qualitative behavior established in the generic $F = 1/F = 2$ -split configuration above, confirming that the optimized EIT cooling scheme identified through the generic parameter scans remains effective once the specific trap parameters of our setup are substituted in place of the illustrative values used during the optimization stage.